ML Assignment 2

Breast Cancer Classification and Streamlit Demonstration

Submission status: Project code, trained models, test data, README, and evaluation results are included. The GitHub URL, deployed Streamlit URL, and BITS Virtual Lab screenshot must be added by the student before final submission.

1. Problem statement

Build, evaluate, and demonstrate multiple classification models on one public classification dataset. The solution must expose the trained models through an interactive Streamlit application that accepts test data, displays evaluation metrics, and shows a confusion matrix and classification report.

2. Dataset description

This project uses the Breast Cancer Wisconsin (Diagnostic) dataset, distributed through scikit-learn and originally published by the UCI Machine Learning Repository. It contains 569 instances, 30 real-valued features, and a binary target: malignant (0) or benign (1). The included test_data.csv contains the stratified 20% hold-out test split (114 rows), generated with random_state=42.

Source: https://archive.ics.uci.edu/dataset/17/breast+cancer+wisconsin+diagnostic

3. Required submission links

Item	Value
GitHub repository	ADD YOUR GITHUB REPOSITORY URL
Live Streamlit app	ADD YOUR STREAMLIT COMMUNITY CLOUD URL

4. Models and evaluation metrics

All models use the same stratified train/test split. Logistic Regression and kNN use standardization; the tree-based models use the original feature values.

ML Model	Accuracy	AUC	Precision	Recall	F1	MCC
Logistic Regression	0.9825	0.9954	0.9861	0.9861	0.9861	0.9623
Decision Tree	0.9211	0.9163	0.9565	0.9167	0.9362	0.8341
kNN	0.9737	0.9884	0.9600	1.0000	0.9796	0.9442
Naive Bayes	0.9386	0.9878	0.9452	0.9583	0.9517	0.8676
Random Forest (Ensemble)	0.9474	0.9937	0.9583	0.9583	0.9583	0.8869

5. Observations

Model	Observation
Logistic Regression	Best overall balance in this run; highest accuracy, F1, and MCC.
Decision Tree	Interpretable but less accurate and less stable than the other models.
kNN	Strong after scaling; perfect recall on this hold-out split, but slightly below Logistic Regression overall.
Naive Bayes	Fast and competitive despite correlated features violating its independence assumption.

Random Forest	Excellent AUC and more robust than one tree, but not the best on this fixed split.
Overall winner	**Logistic Regression**

6. Streamlit application features

- CSV test-data upload with schema validation
- Model-selection dropdown for all five models
- Accuracy, AUC, precision, recall, F1, and MCC display when target is present
- Confusion matrix and classification report
- Predictions table and downloadable predictions CSV

7. Repository structure

```
ml_assignment_2/
■■■ app.py
■■■ requirements.txt
■■■ README.md
■■■ test_data.csv
■■■ model/
    ■■■ model_training.py
    ■■■ logistic_regression.joblib
    ■■■ decision_tree.joblib
    ■■■ knn.joblib
    ■■■ naive_bayes.joblib
    ■■■ random_forest.joblib
    ■■■ metrics.json
```

8. BITS Virtual Lab screenshot

Insert one screenshot showing this assignment running in the BITS Virtual Lab in the area below. This placeholder is intentionally not fabricated.

PASTE BITS VIRTUAL LAB EXECUTION SCREENSHOT HERE

Files included with this submission package: app.py, requirements.txt, README.md, test_data.csv, model/model_training.py, trained model artifacts, and metrics.json.